

Enhancing Anomaly Alert Prioritization Through Calibrated Standard Deviation Uncertainty Estimation With an Ensemble of Auto-Encoders

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Abstract—Deep auto-encoders (AEs) are widely employed deep learning methods in the field of anomaly detection across diverse domains (e.g., cybersecurity analysts managing large volumes of alerts, or medical practitioners monitoring irregular patient signals). In such contexts, practitioners often face challenges of scale and limited processing resources. To cope, strategies such as false positive reduction, human-in-the-loop review, and alert prioritization are commonly adopted. This paper explores the integration of uncertainty quantification (UQ) methods into alert prioritization for anomaly detection using ensembles of AEs. UQ models highlight doubtful classification decisions, enabling analysts to address the most certain alerts first, since higher certainty typically correlates with greater accuracy. Our study reveals a nuanced issue where applying UQ to ensembles of AEs can produce skewed distributions of large reconstruction errors (errors exceeding a pre-defined threshold), which may falsely suggest high uncertainty when standard deviation is used as the metric. Conventionally, a high standard deviation indicates high uncertainty. However, contrary to intuition, large reconstruction errors often reflect AE is strongly confident that an input is anomalous—not uncertainty about it. Moreover, ensembles of AEs generate reconstruction errors with varying ranges, complicating interpretation. To address this, we propose an extension that calibrates the standard deviation distribution of uncertainties, mitigating erroneous prioritization. Evaluation on 10 benchmark datasets demonstrates that our calibration approach improves the effectiveness of UQ methods in prioritizing alerts, while maintaining favorable trade-offs across other key performance metrics.

Index Terms—Auto-encoders, security anomaly detection, alert prioritization, uncertainty estimation.

I. INTRODUCTION

IN THE rapidly evolving landscape of system monitoring, analysts are confronted with the formidable task of navigating through an overwhelming influx of alerts [41], a challenge exacerbated by the constraints posed by limited processing resources. The sheer volume of alerts, often a consequence

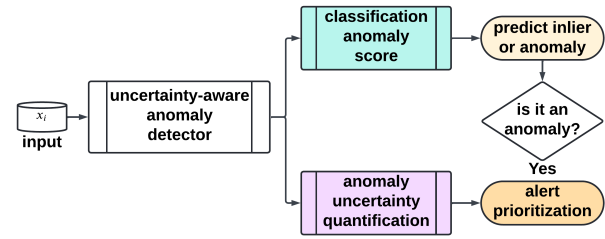


Fig. 1. An uncertainty-aware anomaly detector [54].

of the increasing complexity of threats, demands efficient approaches to alert management [7]. To address this critical issue, analysts employ a range of strategies, among which automated triage, false positive reduction, human-in-the-loop and alert prioritization stand out [57]. These strategies could be used, for instance, to bolster the security of networks (such strategies could also be useful to networks of autonomous robots [27]).

This paper aims to shed light on a crucial aspect of alert management—alert prioritization [2] (which involves determination of the likelihood of an alert to be associated with a genuine security threat). This is particularly pertinent in the realm of intrusion detection (focuses specifically on identifying security threats within a system) or anomaly detection (AD)—identifying deviations from normal behavior, which can include security threats—where distinguishing genuine threats from benign anomalies is paramount. The emphasis here lies on leveraging uncertainty quantification (UQ) methods to enhance the prioritization process. UQ methods play a pivotal role in deciphering the confidence levels associated with classification decisions. By discerning doubtful decisions, UQ models provide cybersecurity analysts with a valuable tool to address the most certain alerts first. This not only optimizes the allocation of resources but also enhances the overall efficacy of the cybersecurity response. Here, a prediction is both a classification decision (as is an instance anomalous?) and an assessment of uncertainty (as is the decision doubtful?).

As shown in Figure 1, UQ and anomaly classification are two parallel processes jointly applied to evaluate an input. Anomaly classification determines whether the input is benign or anomalous. UQ assesses the confidence/uncertainty

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in that classification. Together, these processes provide a more reliable decision-making framework by combining predictive classification with confidence evaluation. In other words, UQ does not alter the classification outcome itself. Rather, it serves as a post-hoc mechanism that evaluates the confidence associated with a decision and supports alert prioritization (i.e., the less uncertain a predicted anomaly, the higher its alert is prioritized). However, if the confidence of a classification decision on an input is truly insufficient—corresponding to very high uncertainty—it could be rejected.

Our research centers on the examination of UQ applied to an ensemble of auto-encoders (AEs)—a particular unsupervised deep learning (DL) architecture—in the context of AD. Our emphasis on AEs is driven by the unique challenges posed by AD systems, where the scarcity of labelled data makes supervised learning approaches impractical. AEs excel in learning intrinsic patterns within the data without the need for labelled data points, making them particularly well-suited for the AD task. This investigation aims to delve into the nuanced interplay between UQ methods and AEs within the AD domain, offering insights into their effectiveness and potential applications in unsupervised scenarios.

Different UQ models exist in literature [1] and have different modelling assumptions and, hence, can yield different uncertainty distributions, depending also on the uncertainty metrics used. Entropy and standard deviation are most commonly used metrics to assess predictive uncertainty (other metrics include Kullback-Leibler divergence, Brier score and log loss), but they capture different aspects of uncertainty, are applicable in various contexts (e.g., intrusion detection systems, finance and manufacturing) and provide complementary insights (the former is often used to measure the impurity of predictions while the latter measures variability of a set of predictions). We observe that ensemble-like UQ methods applied on AEs can produce large uncertainties on large reconstruction errors (relative to a pre-defined classification threshold) when standard deviation is used as uncertainty metric. This occurs when the reconstruction error distribution is skewed. In such cases, using standard deviation as the uncertainty metric means that a high standard deviation indicates high uncertainty. This quantification could be erroneous because, according to AEs, large reconstruction errors could mean small uncertainty i.e., a large reconstruction error often indicates that the model is confident the input is anomalous—not that it is uncertain about it. Also, as an ensemble, different AEs produce reconstruction errors with varying ranges.

Uncertainty can be divided into two main types: aleatoric and epistemic. Standard deviation, our UQ metric, is effective at quantifying epistemic uncertainty. In other words, an ensemble of AEs using standard deviation across reconstruction errors primarily captures epistemic uncertainty (model uncertainty)—i.e., variability in predictions due to differences among ensemble members [8], [26].

What we aim to convey is that when standard deviation is used as an uncertainty metric, the interpretation is straightforward: a high standard deviation implies high uncertainty, while a low standard deviation implies low uncertainty. However, when using an ensemble of AEs, the reconstruction

errors for a given data point often exhibit skewness. This skewness can cause the standard deviation of the reconstruction errors to become artificially large—exceeding the chosen threshold—and thus disproportionately influence the final decision for that data point. Our goal is to mitigate the impact of this skewness when evaluating standard deviation for uncertainty quantification.

An AE model that is uncertain about a classification decision is more likely to mislabel a sample. It should be noted that classification and UQ are distinct tasks here. During prediction, each AE-based model, independently trained, produces a reconstruction error (an AE operates by attempting to reconstruct an input x in minimizing the reconstruction error as much as possible). Then a class is determined by comparing the reconstruction error to a classification threshold, to indicate whether the underlying data point is anomalous or not (0 for normality and 1 for abnormality). The combined classification decisions from all individual predicted classes, using majority vote [28], are considered to classify the underlying data point, and individual reconstruction errors are considered to calculate the uncertainty associated with the classification decision. In other words, classification determines whether a data point is normal or anomalous, while uncertainty quantification assesses whether the classification decision is doubtful. These two are closely related: for example, an anomalous data point that is mistakenly classified as normal should ideally be assigned a high uncertainty score.

Our work makes the following **contributions**:

- **Calibration method.** We identified an interpretation issue in ensembles of AEs when using standard deviation for uncertainty quantification. Importantly, large reconstruction errors (i.e., errors exceeding a pre-defined threshold) do not necessarily correspond to high uncertainty. In practice, a large error often indicates that the AE is confident the input lies outside the distribution of benign data [4], [48]—that is, it is anomalous—rather than uncertain about its prediction. We then propose a technique and an algorithm to calibrate uncertainty distributions from ensembles of AEs for anomaly detection when using standard deviation as the uncertainty metric.
- **Alert prioritization strategy.** We introduce a novel association between low uncertainty and high reconstruction error, enabling more effective prioritization of alerts. We argue that uncertainty quantification can be combined with contextual alert features (e.g., attack patterns, severity, threat source) to support practical prioritization.
- **Empirical evaluation.** We validate the approach on non-temporal/tabular anomaly detection datasets, demonstrating its utility in anomaly detection.

Scope and limitations. Our method builds on existing standard-deviation-based UQ; it is not a new UQ framework. We do not aim to produce fully reliable uncertainty distributions (cf. [54]), and our focus on tabular datasets may limit generalizability to sequential or temporal domains.

While our approach shares similarities with [54]—primarily because ensembles of models can be given a Bayesian interpretation, albeit with certain caveats [14], [21]—we emphasize three key differences:

- **Different uncertainty objective.** [54] employs Bayesian AEs to model total uncertainty (epistemic and aleatoric) by converting negative log-likelihood estimates as uncertainty metric and uses this to reject predictions. In contrast, we focus on ensembles of deep AEs using standard deviation as uncertainty metric and design a calibration procedure that aligns standard-deviation-based uncertainty with reconstruction error, enabling alert prioritization rather than rejection. This shift addresses a distinct operational need in alert monitoring, where analysts must triage alerts under resource constraints rather than discard them.
- **Calibration technique.** We tackle the problem of misleading uncertainty estimates—where large reconstruction errors can spuriously inflate uncertainty—by introducing a calibration strategy that produces more reliable uncertainty scores. This methodological contribution is not present in [54] (indeed, [54] is a UQ method, not a calibration method) and directly improves the utility of ensemble-based UQ in practice.
- **Practical efficiency.** “Pure” Bayesian approaches are computationally demanding, limiting their deployment in real-time anomaly detection pipelines. Our ensemble-based method offers a lighter-weight calibration method that scales better to operational environments while still providing actionable uncertainty information.

It is important to note that the problem tackled here (i.e., calibrating UQ from an ensemble of AEs using standard deviation as the metric) is unprecedented and represents a unique research challenge with no established prior work.

II. RELATED WORK AND BACKGROUND

A. Related Work

Anomaly detection (AD) consists of examining specific data points and detecting occurrences that deviate from the established pattern of behaviors. AD is not new and the first solutions were manual. But as data increases very rapidly manual tracking becomes almost impractical. Various automated AD methods have been proposed in literature, from classical to DL approaches [36] (DL for feature extraction with pre-trained models [49] such as ResNet, VGG and AlexNet composed with a one-class SVM [15]; learning feature representation of normality such as AEs [23] and its variants such as DAGMM [58] and DUAD [23], GANs [43] and predictability modelling-based methods [46]; and end-to-end anomaly score learning such as LSTM [24]). Furthermore, several issues have been considered, dictating trends of contributions in AD. For example, using AEs, models are trained on normal data points only. However, because of constraints of reality, the training data set can be contaminated, i.e., it can contain abnormal data points, unbeknownst to users. This work considers uncontaminated training data sets. Though the methods above are the main baselines, this work considers vanilla AEs only. We consider AEs because AE-based models have consistently demonstrated good results in recent development of AD systems [12], [39].

Alert prioritization. The prioritization of alerts is crucial to effectively manage resources and respond to potential threats.

Several methods have been proposed to address this challenge, each offering unique approaches and considerations. One common approach is risk-based prioritization, which involves assessing the potential impact and likelihood of system alerts. For instance in cybersecurity, by considering factors such as the asset value, threat severity, and exploitability, security teams can prioritize alerts based on their overall risk to the organization. Incorporating threat intelligence feeds can enhance alert prioritization by providing contextual information about emerging threats and attack patterns. By integrating external sources of threat intelligence into the prioritization process, security teams can better understand the relevance and potential impact of security alerts [25]. Some approaches focus on automating the response to security alerts, thereby reducing the burden on human analysts and accelerating incident response times. Automated response mechanisms can range from pre-defined playbooks to dynamic decision-making based on real-time analysis of alert data [31]. DL techniques have also been employed for system alert prioritization. These methods utilize algorithms to analyze historical data, identify patterns, and predict the significance of system alerts [3], [55]. This study focuses on DL techniques to quantify uncertainties for alert prioritization.

Uncertainty quantification (UQ). Various UQ techniques exist to quantify uncertainties (e.g. Bayesian neural networks [16], Variational AEs (VAEs) [18] and ensemble DL methods (EDL) [26], [53]). Bayesian neural networks and VAEs treat model parameters or architecture as random variables and place prior distributions over them, and learn their posterior distribution. It is important to note that UQ techniques yield different uncertainty profiles due to differing modeling assumptions. We focus on ensemble-like setups only for uncertainty quantification (e.g., multiple runs of Monte Carlo dropout (MCD) [11] or VAEs, and EDL) as they provide straightforward interpretability and valuable practical insights—most ensemble-like UQ methods have Bayesian interpretation. However, there are alternative standalone methods in literature for providing uncertainty estimations (e.g., Gaussian processes [35], quantile regression [19] and conformal prediction (CP) [44]). In our experiments, EDL, DAGMM [58], MCD and VAEs (using AEs) will be employed—chosen specifically for their ability to leverage standard deviation. Our study reveals a nuanced issue where applying UQ to an ensemble of AEs can lead to skewed distributions of large reconstruction errors, falsely indicating large uncertainties on certain classification decisions, contrary to the functioning of AEs (with AEs, large reconstruction errors indicate that the model is confident the input is anomalous). We aim to develop a calibration method to scale standard deviation uncertainties falsely indicating large uncertainties.

Calibration methods consist of correcting outcome distribution of DL methods, especially, when the (non-calibrated) outcomes might not align with some underlying properties [38]. Existing calibration methods for UQ consider classification task [34]. The same concepts can be applied to unsupervised task. Producing calibrated uncertainty distribution is also important for model interpretability, i.e., the reliability of a DL model’s confidence in its decisions is

critical for high-risk applications, for example AD systems. Accurate uncertainty estimates provide a valuable extra bit of information to establish trustful AD systems [13]. For classification tasks, calibration is that a model's predicted probabilities of outcomes reflect true probabilities of those outcomes [33].

B. Background

1) *Problem Setting*: Let X denote a data set given by

$$X = X^- \cup X^+ = \{x_i\}_{i=1}^N, \quad (1)$$

where X^- and X^+ are the sets of normal and abnormal data points respectively. Every x_i in X has its label $y_i \in \{0, 1\}$ ($y_i = 0$ for $x_i \in X^-$ and $y_i = 1$ otherwise). Our setting is as follows. An AE model is trained on X^- only. So, the model attempts to minimize a loss function (which represents the sum of reconstruction error e_i on each data point x_i versus its reconstructed copy $\hat{x}_i$) that, using an ℓ_2 -norm for instance (other loss metrics could be used), is

$$e_i = \|x_i - \hat{x}_i\|. \quad (2)$$

Let $\Delta = \Delta^- \cup \Delta^+$ be the set of reconstruction errors where $\Delta^- = \{e_i : x_i \in X^-\}$ and $\Delta^+ = \{e_i : x_i \in X^+\}$.

2) *Auto-Encoder (AE)*: An AE is made of two neural networks, an encoder, f_θ , and a decoder, g_ϕ (the full model is denoted by h_Θ). An encoder consists of learning a latent space of input space. Once a latent space is learned, a decoder tries to reconstruct the input instance x_i from its latent representation z_i [56]. The goal is then to minimize reconstruction error given in (3) by searching for optimal model parameters θ and ϕ , that best reproduce input instances. It should be noted that since UQ models used here are ensemble-like approaches where uncertainties are computed using multiple runs of a model (for MCD or VAEs) or multiple AEs (for EDL), our previous notations would slightly change to annotate the run of a model. A total of K runs would be considered for each UQ method. The notations f_θ will then be replaced by $f_\theta^{(k)}$ to indicate the encoder of the k th run of an AE, g_ϕ by $g_\phi^{(k)}$, h_Θ by $h_\Theta^{(k)}$, z_i by $z_i^{(k)}$, $\hat{x}_i$ by $\hat{x}_i^{(k)}$, e_i by $e_i^{(k)}$ and $\hat{y}_i$ by $\hat{y}_i^{(k)}$.

The objective function of $h_\Theta^{(k)}$ is given by

$$\min_{\Theta} \sum_{x_i \in X^-} e_i^{(k)}, \quad (3)$$

where $e_i^{(k)}$ is defined in (2) and $\Theta = (\theta, \phi)$ denote model's parameter. From what precedes, it is evident that

$$\hat{x}_i^{(k)} = g_\phi(z_i^{(k)}) = g_\phi^{(k)}(f_\theta^{(k)}(x_i^{(k)})). \quad (4)$$

3) *Deep Autoencoding Gaussian Mixture Model (DAGMM)*: is a DL anomaly detection algorithm that combines the strengths of AE-based feature learning and Gaussian mixture models (GMMs) for unsupervised learning tasks [58]. DAGMM is designed to detect anomalies in high-dimensional data, such as images, time-series, or tabular data. It utilizes an AE network to learn a low-dimensional representation of the input data. By training the AE on normal data samples, DAGMM learns to capture the underlying characteristics of the normal data distribution.

Once the AE is trained, DAGMM uses the latent representations learned by the encoder to fit a GMM to the data. The GMM models the distribution of the latent representations as a mixture of Gaussian distributions, where each component represents a cluster of data points with similar characteristics. DAGMM optimizes the parameters of the GMM to maximize the likelihood of the latent space under the mixture model.

After training AE and fitting GMM, anomalies are identified as data points with small likelihood under the learned GMM. Data points that deviate significantly from the normal data distribution in the latent space are considered anomalies.

4) *Ensemble Deep Learning (EDL)*: refers to a technique where multiple DL models (AEs for in this case), often of the same type or architecture, are trained and combined to improve performance [53]. Each individual model in the ensemble might be trained with different weight and bias initializations, subsets of the training data, or hyperparameters, leading to a diverse set of models. The combination of classification decisions from multiple models can often result in better performance in terms of accuracy, generalization, and robustness compared to a single model. Decisions from individual classifiers could be weighted, based on their performance during training (or using prior knowledge), before fusion [29]. Distinction between reconstruction errors from the K models is used to assess uncertainty. Common methods for combining classification decisions in EDL include voting [28] or using more sophisticated techniques such as stacking or boosting [42]. To classify a data point using the k th model $h_\Theta^{(k)}$, a threshold $\eta^{(k)}$ is set during the training regime for inference. Ideally, it should be given as in (5). A new data point x_i is flagged as anomalous if $e_i^{(k)} > \eta^{(k)}$. Otherwise, it is normal. A theoretical threshold for AEs is given by

$$\eta^{(k)} = \max \{e_i^{(k)} : \forall x_i \in X^-\}. \quad (5)$$

In practice, a validation data set is used to determine the optimal thresholds, which is the approach we adopt here. It should be noted that the detection of doubtful decisions based on uncertainty can be sensitive to the choice of uncertainty threshold [54].

5) *Variational AEs (VAEs)*: are a type of AEs and are trained using variational inference, where the model learns to approximate the true data distribution by optimizing a lower bound on the log-likelihood of the data [18]. The key difference between classical AEs and VAEs is in how they represent and learn the latent space. VAEs introduce a probabilistic interpretation of the latent space and use variational inference to train the model, allowing them to capture uncertainty and generate more diverse and meaningful representations compared to classical AEs. The objective function for training VAEs typically consists of two terms: the reconstruction loss, which measures the fidelity of the reconstructed data, and the Kullback-Leibler (KL) divergence between the learned latent distribution and a predefined prior distribution. Mathematically, the objective function for training VAEs can be expressed as:

$$\mathcal{L}_{\text{VAE}} = -\mathbb{E}_{q(z|x)}[\log p(x|z)] + D_{\text{KL}}(q(z|x)||p(z)), \quad (6)$$

where $q(z|x)$ denotes the approximate posterior distribution over latent variables given input data x , $p(x|z)$ represents the likelihood of generating data x given latent variables z , and $p(z)$ represents the prior distribution over latent variables. D_{KL} denotes the KL divergence, given by

$$D_{\text{KL}}(P||Q) = \int P(x) \log \frac{P(x)}{Q(x)} dx \quad (7)$$

At each run of the same trained VAE, a different reconstruction error could be obtained. This is what makes an ensemble-like predictions from which uncertainty is quantified.

6) *Monte Carlo Dropout (MCD)*: is a method for estimating uncertainty in deep neural networks (AEs in this case) by applying dropout at inference time and performing Monte Carlo sampling [11]. Dropout is a regularization technique commonly used during training to prevent overfitting. During training, dropout randomly sets a fraction of the neurons to zero, effectively removing them from the network. At inference time, MCD applies dropout stochastically and performs multiple forward passes through the network with different dropout masks. It then computes statistics, such as the mean and variance, of the predictions across these forward passes to estimate uncertainty.

III. OUR PROPOSED APPROACH: MOSEMBOLI

We propose a standard deviation uncertainty quantification, named Mosemboli¹ (whose architecture is shown in Figure 2b), to enhance expression of uncertainty distribution obtained from existing UQ methods. Similarly to [54], our proposed calibration method is built on top of a UQ method—its inputs are uncertainty distribution obtained from existing UQ methods. It is important to note that our proposed method does not change classification decisions, but uncertainty distributions only. Our observation is as follows: since an AE is a threshold-based method, it should be less uncertain on its anomalous classification decisions with large reconstruction errors where all K classification decisions converge (i.e., all models agree on the nature of the underlying data point)—this is a fundamental principle of AEs; otherwise, the model should have large uncertainty (the same observation was made in [54] but with entropy alone as metric where they summarized the anomaly probabilities from all Bayesian AEs posterior samples via expectation, so standard deviation was not considered)—the deviation between a reconstruction error and a fixed threshold is used to enhance the expression of uncertainty distribution obtained from UQ methods. Hence, the proposed calibration method, inspired by the Gaussian distribution mixture method [40], consists of arranging reconstruction errors into 3 segments (see Figure 2b), namely: normality, abnormality and uncertainty zones, where uncertainties in reconstructing inputs are then expressed in terms of how the three segments or distributions overlap when plotting the reconstruction errors.

It was indicated that it is not trivial to determine a threshold using AEs. In practice, an optimal value of threshold using a

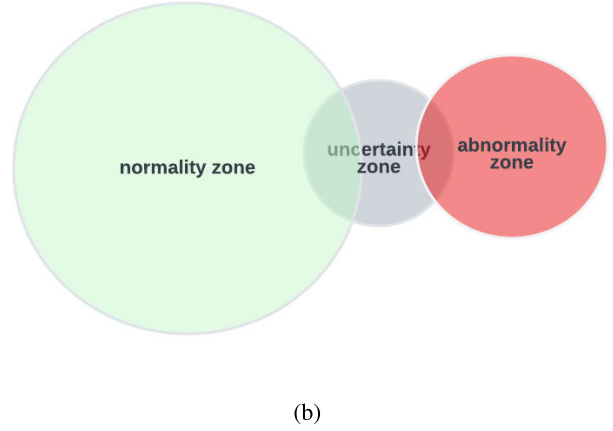
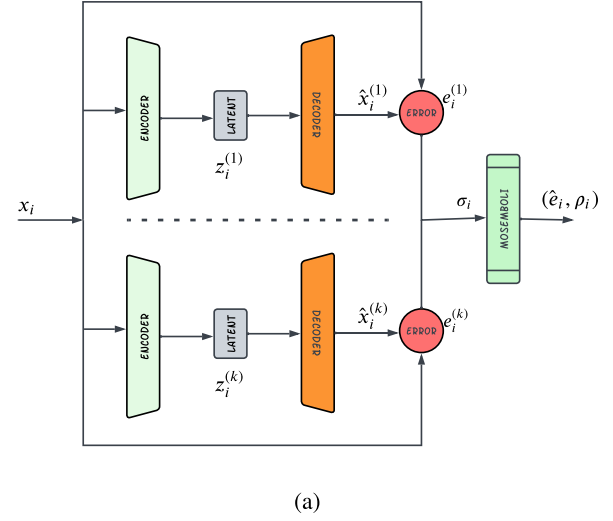


Fig. 2. Mosemboli: (2a) proposed architecture and (2b) an illustration of reconstruction error distribution mixture and how adjacent zones can overlap to determine the scaling factor.

validation set is found in the neighborhood of $\eta^{(k)}$ defined in (5). Here, the zone where a threshold could be found is viewed as an uncertainty zone. This means that, given an instance x_i , its uncertainty will be large if its reconstruction error is found to be in the uncertainty zone. Conversely, an AE model should have relatively small uncertainty for reconstructions where errors are outside of the uncertainty zone. Thus, there would be three distributions for reconstruction errors on normal, anomalous and uncertain data points where each implements an uncertainty distribution in a subpopulation of the error reconstruction space. The 3 distributions should be combined to produce a calibrated uncertainty distribution. Before delving further, we introduce some properties that Mosemboli satisfies.

A. Properties of the Calibration Method

Given K trained versions of a UQ model (see Figure 2a), each version $h_{\Theta}^{(k)}$ produces a reconstruction error $e_i^{(k)}$ on data instance x_i . Here, a model's version means a run from the probabilistic UQ method. An uncertainty σ_i is the standard

¹Mosemboli means a calibrator in Lingala, a local language spoken in the Democratic Republic of Congo.

deviation given by

$$\sigma_i = \sqrt{\sum_{k=1}^K p_i^{(k)} \left(e_i^{(k)} - \mu_i \right)^2} \text{ where } \mu_i = \sum_{k=1}^K p_i^{(k)} e_i^{(k)}, \quad (8)$$

with $p_i^{(k)}$ denoting the probability associated with $e_i^{(k)}$, given, for instance, by

$$p_i^{(k)} = \frac{e_i^{(k)}}{s_i}, \quad (9)$$

where

$$s_i = \sum_{k=1}^K e_i^{(k)}.$$

Equation (8) is what a UQ method should produce as measures of uncertainty. Uncertainty distribution should be calibrated only for situations where classification decisions converge (all members of the ensemble are in agreement on the nature of the underlying data point), i.e., only data points x_i associated with reconstruction errors $e_i^{(k)}$ satisfying the following conditions: for normal data

$$e_i^{(k)} < \eta^{(k)} \quad \forall k, \quad (10)$$

and for anomalous data

$$e_i^{(k)} \geq \eta^{(k)} \quad \forall k. \quad (11)$$

Equations 10 and 11 are a form of normalization of reconstruction errors (a form of agreement in classification decisions) from various models, and need to hold on samples which uncertainties require calibration. r_i^- , r_i^+ and r_i^* denote the probability that the mean reconstruction error $\hat{e}_i$ of a data point x_i be a member of normal, abnormal or uncertain zones respectively. The calibrated uncertainty ρ_i on the average $\hat{e}_i$ is

$$\rho_i = \frac{r_i^*}{r_i^- + r_i^+ + r_i^*} \times \sigma_i, \quad (12)$$

where r_i^* can be viewed as a scaling factor [38].

An AE-based calibration method to enhance uncertainty distribution should satisfy 3 properties, which are taken from the basic functioning principle of AEs: expected behavior of a system is drawn from the same learnable data distribution (i.e., normality) while unexpected behavior is difficult to learn.

- **(normality)**: the more similar the reconstructed version of an input is to the input the less uncertain the model should be. Mathematically, for $\hat{e}_i \geq \hat{e}_j$ and $\hat{e}_i \leq \eta$,

$$r_i^- \leq r_j^-. \quad (13)$$

Equation (13) indicates the less e_i gets the smaller $r^*(e_i)$ gets, i.e., this better aligns with expected behavior of AEs on normal data as the less reconstruction errors get the higher is the chance that the instance is normal.

- **(abnormality)**: the more dissimilar the reconstructed copy of an input is to the input the less uncertain the model should be. Mathematically, for $\hat{e}_i \leq \hat{e}_j$ and $\hat{e}_i \geq \eta$,

$$r_i^+ \geq r_j^+. \quad (14)$$

Equation (14) indicates the larger $\hat{e}_i$ gets the smaller r_i^* gets, i.e., this better aligns with the behavior of AEs on

Algorithm 1 Mosemboli

```

1: function MOSEMBOLI( $h_\Theta^{(1)}, \dots, h_\Theta^{(K)}, \eta, \varphi, d, X_{\text{val}}$ )
2:    $E \leftarrow []$ 
3:   for  $x_i \in X_{\text{val}}$  do
4:      $e_i \leftarrow 0$ 
5:     for  $h_\Theta^{(k)} \in \{h_\Theta^{(1)}, \dots, h_\Theta^{(K)}\}$  do
6:        $e_i \leftarrow e_i + \|x_i - h_\Theta^{(k)}(x_i)\|$ 
7:        $E.\text{push}(e_i)$ 
8:     end for
9:      $\delta^- \leftarrow (\eta - \min(E))d$ 
10:     $\delta^+ \leftarrow (\max(E) - \eta)d$ 
11:  end for
12:   $(\Gamma^-, \Gamma^+, \Gamma^*) \leftarrow (\emptyset, \emptyset, \emptyset)$ 
13:  for  $x_i \in X_{\text{val}}$  do
14:    if  $e_i < \eta - (1 - \varphi)\delta^-$  then
15:       $\Gamma^- \leftarrow \Gamma^- \cup \{x_i\}$ 
16:    else if  $e_i > \eta + (1 - \varphi)\delta^+$  then
17:       $\Gamma^+ \leftarrow \Gamma^+ \cup \{x_i\}$ 
18:    else
19:       $\Gamma^* \leftarrow \Gamma^* \cup \{x_i\}$ 
20:    end if
21:  end for
22:   $\text{fit}(D^-, \Gamma^-)$ 
23:   $\text{fit}(D^+, \Gamma^+)$ 
24:   $\text{fit}(D^*, \Gamma^*)$ 
25:  return  $(D^-, D^+, D^*)$ 
26: end function

```

anomalous data as the larger reconstruction errors get the larger is the chance that the instance is anomalous.

- **(symmetry)**: there is a zone where the AE model is indecisive on the relation between the reconstructed version of the input and the input itself, i.e., the distribution in this zone is symmetric. Mathematically, $\exists \varepsilon > 0$ such that

$$r_{\eta-\varepsilon}^* = r_{\eta+\varepsilon}^*. \quad (15)$$

Equation (15) indicates that the distribution curve in the uncertainty area can be divided around the AE threshold η to produce two equal halves where the model finds it difficult to distinguish categories of instances.

Mosemboli should satisfy these 3 properties. However, according to (12), this might not be the case as the value of ρ_i depends also on σ_i . This means that for some values of σ_i , (e.g., $\sigma_i = 0$), ρ_i might not be regular.

B. Proposed Algorithm

To quantify a scaling factor r_i^* the following are considered: reconstruction errors on (a) normal data points follow a distribution D^- , (b) abnormal data points follow another distribution D^+ , and (c) uncertain data points follow another distribution D^* . It is intended to design an uncertainty quantification method that will indicate large uncertainty when reconstruction errors fall on the uncertainty zone.

Let π_i^- denote a probability that a data point x_i is normal (i.e., the larger π_i^- gets the more chance the underlying data point is normal), π_i^+ denote a probability that a data point x_i

Algorithm 2 An Example of How to Use Mosemboli

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1: procedure MAIN( $h_{\Theta}^{(1)}, \dots, h_{\Theta}^{(K)}, \eta, \varphi, d, X_{\text{val}}, X_{\text{test}}$ )
2:    $(D^-, D^+, D^*) \leftarrow \text{MOSEMBOLI}(h_{\Theta}^{(1)}, \dots, h_{\Theta}^{(K)}, \eta, \varphi, d, X_{\text{val}})$ 
3:   for  $x_i \in X_{\text{test}}$  do
4:      $(e_i, \sigma_i) \leftarrow \text{BASELINE-UQ}(h_{\Theta}^{(1)}, \dots, h_{\Theta}^{(K)}, x_i)$ 
5:      $\pi_i^- \sim D^-(e_i)$ 
6:      $\pi_i^+ \sim D^+(e_i)$ 
7:      $\pi_i^* \sim D^*(e_i)$ 
8:      $\rho_i \leftarrow \frac{\pi_i^*}{\pi_i^- + \pi_i^+ + \pi_i^*} \sigma_i$ 
9:     print( $\sigma_i, \rho_i$ )
10:  end for
11: end procedure

```

is anomalous, and π_i^* denote a probability that a data point x_i is uncertain. The mixture of the 3 distributions is

$$\pi_i = \frac{1}{\pi_i^- + \pi_i^+ + \pi_i^*} \begin{pmatrix} \pi_i^- \\ \pi_i^+ \\ \pi_i^* \end{pmatrix}. \quad (16)$$

The calibrated uncertainty ρ_i on an instance x_i is

$$\rho_i = \frac{\pi_i^*}{\pi_i^- + \pi_i^+ + \pi_i^*} \times \sigma_i. \quad (17)$$

Thus, as shown in Figure 2a, the proposed method outputs two scaled values $\hat{e}_i$ and ρ_i which denote the average error reconstruction from versions of a UQ method and the calibrated standard deviation respectively. Our resulting algorithm is detailed in Algorithm 1 (and an example of how it can be used is given in Algorithm 2). The code is available at <https://anonymous.4open.science/r/ddm-8D63/>.

C. On the Implementation of the Proposed Algorithm

Recall that the proposal considers already trained models. Models are trained on normal data points only. Further, calibration requires parameters for the three reconstruction error distributions (we use exponential, Gaussian and cumulative exponential distributions). The method to get distributions' parameters to measure the inter-zone overlap is through reconstruction errors of a held-out validation data set X_{val} . We need to estimate six parameters (two for each of the three distributions), namely: (μ^-, α^-) for the exponential distribution, (μ^+, α^+) for the cumulative exponential distribution and (μ^*, α^*) for the Gaussian distribution. Let d^- and d^+ denote the width of reconstruction errors on the left and right sides of η (the value of η is determined using a held-out validation dataset) respectively, given by

$$d^- = \eta - \min_{x_i \in X_{\text{val}}} e_i \text{ and } d^+ = \max_{x_i \in X_{\text{val}}} e_i - \eta. \quad (18)$$

Let d denote the radius of the uncertainty zone. We define the following measures,

$$\delta^- = d \times d^- \text{ and } \delta^+ = d \times d^+. \quad (19)$$

We set the three reconstruction error samples as follows

$$\Gamma^- = \{x_i \in X_{\text{val}} | e_i < \eta - (1 - \varphi)\delta^-\}, \quad (20)$$

corresponding to normal data points,

$$\Gamma^+ = \{x_i \in X_{\text{val}} | e_i > \eta + (1 - \varphi)\delta^+\}, \quad (21)$$

to anomalous data points and

$$\Gamma^* = \{x_i \in X_{\text{val}} | \eta - (1 + \varphi)\delta^- \leq e_i \leq \eta + (1 + \varphi)\delta^+\}, \quad (22)$$

to data points on the uncertainty zone where φ denotes some level of overlapping between every pair of direct adjacent distributions (i.e., overlaps between distributions of normality and symmetry, and between symmetry and abnormality). Here, we set reconstruction errors in Γ^- to follow the exponential distribution with parameters (μ^-, α^-) , in Γ^+ to follow the cumulative exponential distribution with parameters (μ^+, α^+) and in Γ^* to follow the Gaussian distribution with parameters (μ^*, α^*) . We fit the three sets Γ^- in (20), Γ^+ in (21) and Γ^* in (22) to estimate distribution parameters using SciPy [51]. Thus, the three probability distributions π_i^- , π_i^+ and π_i^* defined in (16) are then densities in Γ^- , Γ^+ and Γ^* respectively. We mainly use $d = 0.3$ and $\varphi = 0.4$ throughout our experiments (other values could be used—in fact, other values were used and produced similar results, which is desirable—the proposed calibration method is not sensitive to these parameters).

D. Toy Example

To illustrate our proposed method, let's consider the scenario where 5 reconstruction errors (refer to (23)) are generated from 5 AE models on a single data point:

$$e = \begin{pmatrix} 1 \\ 1.2 \\ 1.4 \\ 1.3 \\ 2.5 \end{pmatrix}, \quad (23)$$

with the classification threshold set, for example, to 0.8 for all the 5 models (but each model could have its own classification threshold). All the members exhibit reconstruction errors surpassing the threshold, implying the anomalous nature of the underlying instance. So, the final class of x is $y_x = 1$. We need now to quantify uncertainty associated to this classification decision. To consider a classification decision doubtful, the associated uncertainty should exceed some uncertainty threshold. When using standard deviation as an uncertainty metric with, for example, the uncertainty threshold set to 0.3, the associated uncertainty σ for predicting the data point is approximately 0.6, considerably large in this instance ($0.6 > 0.3$). For this case, despite the counterintuitive nature, as per the functioning of AEs, this large standard deviation implies small uncertainty (that is because individual large reconstruction errors indicate that all five models are confident the underlying data input is anomalous)—the intuitive nature is that a small standard deviation should indicate small uncertainty. We then need a scaling factor κ to rectify this counterintuitive effect. For instance, setting $\kappa = 0.1$ results in a calibrated standard deviation uncertainty of $\kappa\sigma = 0.06$, indicating indeed small uncertainty in this case. This paper revolves around the narrative of calibrating standard deviation uncertainty when employing vanilla AEs in an ensemble fashion.

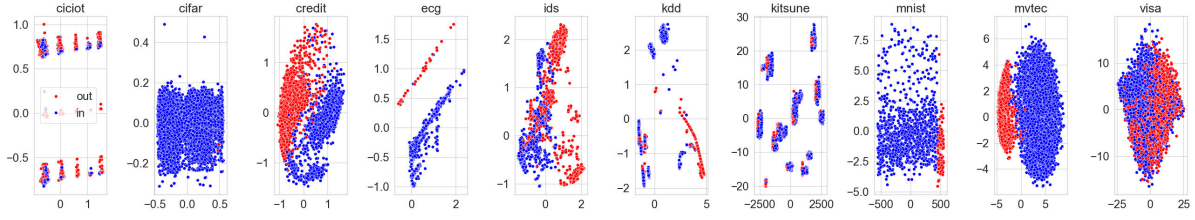


Fig. 3. Visualisation of training data sets.

IV. EXPERIMENTAL INVESTIGATION

A. Experimental Setup

To assess the contribution of the proposed method (Mosemboli) to existing UQ methods (namely, EDL-AE, DAGMM, MCD and VAEs), the following metrics will be used: accuracy (ACC), F_1 -score (F_1), inversion number [9] (IN), root mean square calibration error [33] (RMSCE), correctly rejection rates (CRR), and analysis of variance of uncertainties. We count (in percentage) number of instances where the model correctly rejects misclassified decisions. CRR is used to validate alert prioritization. We vary the uncertainty threshold τ from 10th to [10, 35]th percentile so that only alerts with smaller uncertainties are considered for prioritization—predicted alerts with uncertainties below the threshold are given higher priority than those with higher uncertainties. We use IN to quantify the difference between the predicted ranking and the ground truth ranking of alerts (here, the ground truth ranking of alerts refers to the expected rankings by AEs). An inversion occurs when the order of alerts in the predicted ranking is opposite to their order in the ground truth ranking—a smaller IN suggests that the model is effective in correctly prioritizing alerts. It is important to highlight that accuracy and F_1 -score are computed based on the model's classification decisions, whereas IN, RMSCE, WA, and WR are calculated using the uncertainties associated with the model's reconstruction errors. Further, to form an ensemble for UQ, we consider a set of 5 candidates as suggested by [21] and [34]. To evaluate ACC and F_1 , each k th version of the model estimates its classification decision, $\hat{y}_i^{(k)}$, using its classification threshold and a majority vote is applied to obtain the final flag [28]. Given $\Lambda = \{x_i \in X_{\text{test}} | \sigma_i \geq \tau \wedge \hat{y}_i \neq y_i\}$,

$$\text{CRR} = \frac{\#\Lambda}{\#X_{\text{test}}} . \quad (24)$$

We expect calibrated UQ methods to preserve the same ACC, F_1 and RMSCE (our proposed method does not minimize RMSCE because this metric is meant to measure calibration which targets overconfidence as discussed in literature), to have small IN and large CRR. We use Pytorch [37] to measure RMSCE. For analysis of variances, the p -value and effect size from hypothesis tests (are commonly used as an indication of noteworthy differences between sets of values [10]) are employed. The significance level for p -value is set to 0.01 throughout. We use SciPy [51] to run the Wilcoxon signed-rank and the Friedman tests. We use Kendall's W to measure the effect size ν [50]. Based on Cohen's interpretation guidelines [6], the values of ν are interpreted as follows. For

TABLE I
THE CHARACTERISTICS OF DATA SETS TO TRAIN
UQ METHODS

Data set	# samples	# attributes	untreated anomaly (%)
CICIOT	416985	41	45
CIFAR-10	6000	1024	90
CREDIT	234333	30	0.2
ECG	4998	140	58
IDS	430256	95	43
KITSUNE	210171	116	23
MNIST	7000	784	90
MVTec	5354	1048576	58
Visa	10821	95	11

$\nu \in [0, 0.3)$, the effect size is negligible; for $\nu \in [0.3, 0.5)$, it is medium; and it is large when $\nu \geq 0.5$.

We use these data sets to demonstrate the effectiveness of our approach (some of these data sets are originally designed for anomaly detection, while others are multi-class classification data sets that we adapt into anomaly detection). The distributions of these data sets are visualized in Figure 3, and their characteristics are summarized in Table I: (1) **CICIOT** is a collection of real-time data containing 33 attacks that are executed in an IoT network [32]. (2) **CIFAR-10** is a widely used data set in the field of machine learning and computer vision, consisting of color images across 10 different classes [20]. Classes include airplanes, automobiles, birds, cats, deer, dogs, frogs, horses, rabbits, and ships. (3) **CREDIT** is credit fraud data set comprising financial transaction records annotated with binary labels indicating fraudulent or legitimate transactions [52]. (4) **ECG** consists of recordings of electrical activity of the heart, capturing expected waveform patterns and anomalies [17]. (5) revised **IDS** contains simulated complex network traffics and several types of attacks [45]. (6) revised **KDD** is an intrusion detection data set [47], containing simulated military traffics and several types of attacks. (7) **KITSUNE** is a collection of 9 network attack data sets captured from either an IP-based commercial surveillance system or an IoT network [30]. (8) **MNIST** consists of grayscale images of handwritten digits (0-9) [22]. MNIST and CIFAR-10 are not naturally designed for anomaly detection tasks. To make them anomaly detection data sets, we consider one class (among provided classes) as inliers. (9) **MVTec** is a data set for benchmarking anomaly detection methods with a focus on industrial inspection. It contains high-resolution images divided into fifteen different object and texture categories. Each category comprises a set of defect-free training images and a test set of images with various kinds of defects as well as images without defects [5]. (10) **Visa** is an

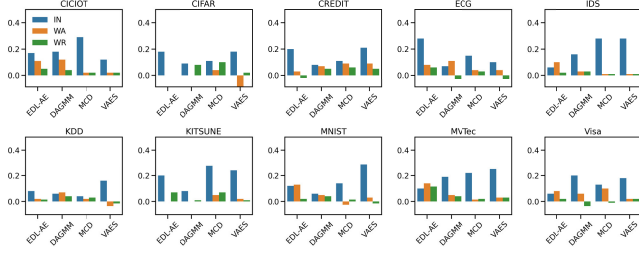


Fig. 4. Differences between uncalibrated and calibrated UQ. Positive values indicate that calibrated UQ are better.

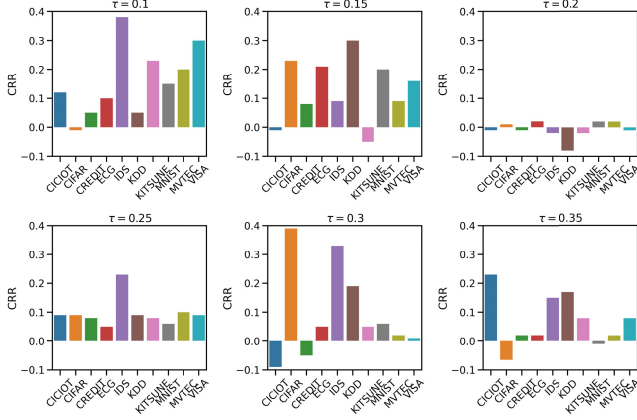


Fig. 5. Average differences in CRR between calibrated and uncalibrated UQ methods. Positive values mean improvement.

anomaly detection data set containing objects from 12 classes of images [59].

While a couple of the chosen data sets may be considered somewhat dated, they still serve as valuable benchmark data sets for anomaly detection methods.

For the notations in various tables and figures, if M denotes an uncalibrated UQ method, $M+$ will denote the calibrated version of the same UQ method.

To evaluate model performance, we use 70% of the benign data for training (i.e., AEs are unsupervised models trained solely on benign data), while the remaining 30% of the benign data together with all anomalous data are reserved for testing. From the test set, 30% is further allocated as a validation set for threshold selection and early-stopping verification. The model hyperparameters are: weight decay of 10^{-5} , batch size of 32, 100 training epochs, patience of 5 for early stopping, learning rate of 10^{-5} , the Adam optimizer, quadratic loss as objective function and the momentum to 0.9 for all AE models. For baselines, we rely on the original published implementations. For models, we use an AE (with 8 layers) and its three variants: MCD, which keeps dropout active at inference for uncertainty; VAEs, which output distributions for stochastic reconstructions; and DAGMM, which augments the AE with a Gaussian mixture network combining reconstruction error and sample energy for anomaly detection.

B. Results and Discussion

Our key finding is that predictions flagged as uncertain must genuinely reflect uncertainty; otherwise, we risk generating

TABLE II
PERFORMANCES: ACC, F_1 , IN AND RMSCE OF THE UNCALIBRATED AND CALIBRATED VERSIONS OF DIFFERENT UQ METHODS

Methods/Metrics	ACC	F ₁	IN	RMSCE	ACC	F ₁	IN	RMSCE
	CICIOT				CIFAR			
EDL-AE	0.65	0.62	0.62	0.25	0.85	0.84	0.60	0.45
EDL-AE+	0.65	0.62	0.45	0.28	0.85	0.84	0.42	0.41
DAGMM	0.80	0.81	0.54	0.24	0.87	0.84	0.54	0.18
DAGMM+	0.80	0.81	0.36	0.28	0.87	0.84	0.44	0.27
MCD	0.70	0.68	0.63	0.24	0.87	0.87	0.46	0.41
MCD+	0.70	0.68	0.34	0.25	0.87	0.87	0.35	0.42
VAEs	0.82	0.82	0.64	0.20	0.89	0.89	0.68	0.44
VAEs+	0.82	0.82	0.52	0.19	0.89	0.89	0.50	0.44
	CREDIT				ECG			
EDL-AE	0.65	0.47	0.61	0.30	0.88	0.90	0.50	0.26
EDL-AE+	0.65	0.47	0.43	0.25	0.88	0.90	0.36	0.21
DAGMM	0.77	0.76	0.57	0.27	0.78	0.79	0.49	0.23
DAGMM+	0.77	0.76	0.46	0.24	0.78	0.79	0.34	0.19
MCD	0.68	0.68	0.60	0.26	0.81	0.82	0.51	0.24
MCD+	0.68	0.68	0.41	0.23	0.81	0.82	0.38	0.20
VAEs	0.89	0.89	0.66	0.25	0.91	0.92	0.52	0.22
VAEs+	0.89	0.89	0.51	0.21	0.91	0.92	0.40	0.18
	IDS				KDD			
EDL-AE	0.59	0.60	0.72	0.38	0.78	0.78	0.53	0.71
EDL-AE+	0.59	0.60	0.65	0.38	0.78	0.78	0.45	0.72
DAGMM	0.64	0.65	0.32	0.70	0.91	0.90	0.42	0.60
DAGMM+	0.64	0.65	0.15	0.73	0.91	0.90	0.35	0.59
MCD	0.60	0.60	0.72	0.39	0.67	0.68	0.55	0.71
MCD+	0.60	0.60	0.43	0.39	0.67	0.68	0.47	0.71
VAEs	0.62	0.62	0.73	0.39	0.91	0.91	0.67	0.72
VAEs+	0.62	0.62	0.41	0.39	0.91	0.91	0.51	0.72
	KITSUNE				MNIST			
EDL-AE	0.75	0.70	0.62	0.43	0.78	0.82	0.44	0.33
EDL-AE+	0.75	0.70	0.42	0.43	0.78	0.82	0.32	0.29
DAGMM	0.76	0.76	0.42	0.63	0.85	0.86	0.38	0.26
EDL-DAGMM+	0.76	0.76	0.33	0.64	0.85	0.86	0.28	0.22
MCD	0.76	0.78	0.64	0.42	0.83	0.83	0.40	0.25
MCD+	0.76	0.78	0.35	0.42	0.83	0.83	0.29	0.22
VAEs	0.73	0.73	0.65	0.28	0.86	0.86	0.37	0.24
VAEs+	0.73	0.73	0.40	0.28	0.86	0.86	0.26	0.20
	MVTec				VISA			
EDL-AE	0.79	0.77	0.51	0.31	0.74	0.75	0.50	0.36
EDL-AE+	0.79	0.77	0.38	0.27	0.74	0.75	0.35	0.32
DAGMM	0.81	0.80	0.44	0.30	0.78	0.76	0.41	0.35
DAGMM+	0.81	0.80	0.33	0.26	0.78	0.76	0.30	0.29
MCD	0.83	0.82	0.48	0.30	0.76	0.78	0.42	0.34
MCD+	0.83	0.82	0.36	0.26	0.76	0.78	0.31	0.28
VAEs	0.84	0.84	0.47	0.27	0.80	0.80	0.43	0.33
VAEs+	0.84	0.84	0.34	0.24	0.80	0.80	0.30	0.27

false alerts. Specifically, if uncertainty is misquantified, it may lead to deprioritizing important events or even causing harm in high-stakes applications. This is particularly critical when the standard deviation is high—we must ensure that it truly indicates high uncertainty. One way to validate this is by checking whether the data point is actually anomalous. In other words, if a data point exhibits high standard deviation but is, in fact, normal, this suggests a misestimation of uncertainty. To assess this issue, we analyze the number of misclassified data points with uncertainty—based on traditional uncertainty quantification methods—that are not anomalous, and conversely, the number of data points with low standard deviations that are anomalous. We expect our method to improve alert prioritization performance under these challenging conditions, while preserving comparable classification performance to traditional methods in typical scenarios.

1) *Improved Uncertainty Quantification:* Figure 5 illustrates that, averaged across datasets and predictive methods, there is a 9% improvement in the rejection of misclassified data points. These correspond to samples that were misclassified but exhibit high uncertainty, primarily driven by skewed reconstruction error distributions. This underscores the importance of uncertainty calibration for downstream tasks

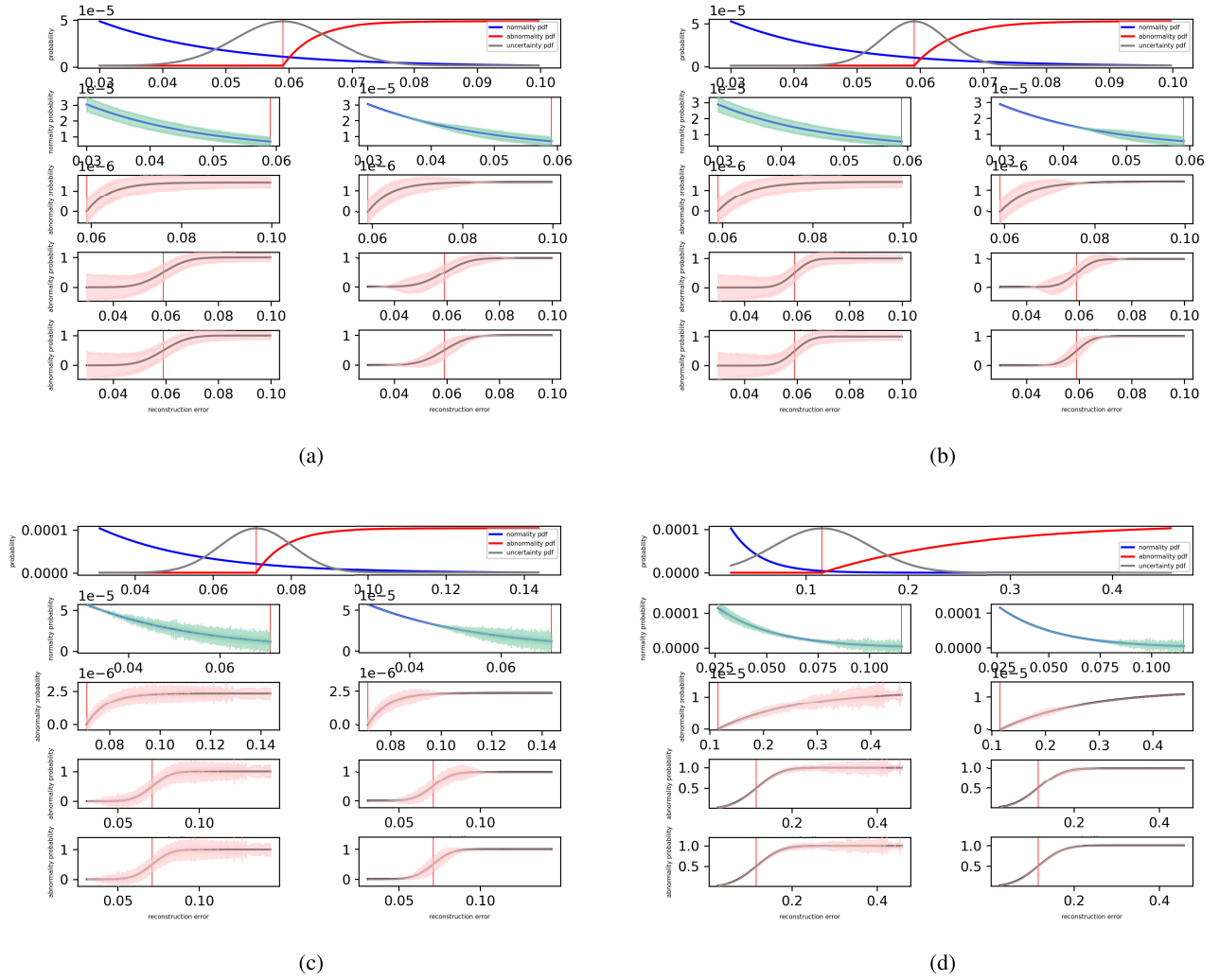


Fig. 6. A sample of uncertainty distributions on KDD ($d = 0.3$ and $\varphi = 0.4$) where EDL-AE (6a), DAGMM (6b), MCD (6c) and VAEs (6d). Blue and red distributions are for normality and abnormality respectively. The top subplot represents the 3 distributions to model the 3 zones. The x-axis denotes reconstruction errors. Green and red shades are levels of uncertainties on reconstruction errors obtained from normal and abnormal instances respectively. Vertical red lines denote the threshold.

such as prioritizing system alerts. Notably, uncalibrated uncertainties can outperform calibrated ones when the ensemble's reconstruction errors do not exhibit skewness, in which case calibration offers limited benefit. UQ methods demonstrate varying effectiveness in identifying doubtful classification decisions. For example, MCD and EDL with calibration yield significantly improved correctly rejection rates (CRR) compared to their uncalibrated counterparts, whereas VAEs show a favorable trade-off. In most cases, calibrated UQ surpasses uncalibrated UQ, indicating that calibration enhances alert prioritization. This indicates that certain high uncertainty estimates—reflected by large standard deviations—can be misleading. Calibrated UQ methods helped mitigate these cases. While CRR highlights the importance of calibration, more robust validation requires reliable ground truth uncertainty labels. For instance, security experts could provide manual assessments of uncertainty associated with specific alerts to better evaluate UQ performance.

2) *Improved Alert Prioritization*: Table II and Figure 4 demonstrate that calibrated UQ methods consistently yield

lower inversion numbers (IN), as expected. This suggests that calibrated UQ methods prioritize alerts more effectively than their uncalibrated counterparts. This means that false high uncertainty can disrupt alert prioritization by drawing attention away from truly ambiguous or critical cases. Lower IN correspond to a stronger agreement between the predicted alert ranking and the expected ranking, indicating greater consistency. Such consistency serves as a robust measure of the model's effectiveness in alert prioritization, thereby increasing system analysts' confidence that the prioritized alerts align with anticipated outcomes. Notably, despite elevated uncertainty stemming from skewed reconstruction errors in certain misclassified samples (as illustrated in Figure 5), our calibration approach successfully mitigates the inconsistency between uncertainty estimates and classification outcomes.

3) *Sustained Classification Performance Reports*: Each calibrated UQ method has exactly the same ACC and F_1 , and almost the same RMSCE than its uncalibrated version across data sets as expected—system analysts can have confidence when utilizing the calibrated uncertainty for alert prioritization

because the calibration process did not impact the overall reliability of classification decisions. Calibration error arises when the observed frequency of true positives deviates consistently from the predicted uncertainties. Although not addressed here, reducing calibration error is crucial for enhancing the reliability of systems.

4) *Statistically Validated Improvements of Our Calibrated Method:* We observed a p -value of less than 0.01 on uncertainties between versions of EDL-AE, DAGMM, MCD and VAEs, and between UQ, UQ+ and all versions of UQ methods. This means that there is a significant difference between every two versions of these approaches. This implies that, because every UQ method and its calibrated version give the same calibration error RMSCE, it appears to prefer the calibrated versions that produce small IN. We also run the Wilcoxon signed-rank test and the Friedman test between the uncalibrated and calibrated versions of UQ methods on their IN. We obtained a p -value of 0.0093 and an effect size of 0.5333, which both indicate that calibrated and uncalibrated UQ methods produce markedly different levels of sortedness of uncertainties.

5) *Increased Complexity Calculation:* Assuming that L is the size of the validation data set X_{val} and K the size of the ensemble, the time complexity of our solution is $\mathcal{O}(KL)$. The time complexity of the baseline UQ method is $\mathcal{O}(K)$. So, using our solution adds a cost on the baseline UQ method, resulting a time complexity of $\mathcal{O}(KL)$. However, for typical L and K , this additional complexity is hardly noticeable.

6) *Distributions of Calibrated Uncertainties:* Figure 6 shows (for illustrative purposes—only results on one data set are shown) reconstruction error and uncertainty distributions using various UQ methods considered here, for $d = 0.3$ and $\varphi = 0.4$. It was observed that different values of d and φ give almost the same results—i.e., the proposed calibration method is not sensitive to the choices of these 2 parameters, which is desirable. In Figure 6d, VAEs have expected uncalibrated and better calibrated expressions of the uncertainty distribution. Compared to other UQ methods, this might mean that VAEs have better ways of expressing uncertainty distributions.

V. CONCLUSION

In this study, we presented a novel method for calibrating standard deviation uncertainty distributions derived from ensemble-based UQ methods applied to AEs in the domain of anomaly detection. Our approach leverages calibration techniques to enhance the prioritization of alerts—a critical aspect of alert management. Particularly, we addressed scenarios where the certainty levels of alerts are expressed through standard deviation, emphasizing that alerts with small standard deviation uncertainties should be considered to be prioritized. Existing ensemble-based UQ methods on AEs, however, often exhibit a counterintuitive assignment of large uncertainty to large-variance reconstruction errors. Our observations led to the introduction of a calibration method designed to rectify this counterintuitive utilization of standard deviation. By aligning uncertainty distribution with reconstruction errors, our method promotes a more effective prioritization strategy for system alerts in anomaly detection scenarios.

There are a couple of avenues of interesting work to further develop our understanding of the uncertainty quantification methods on auto-encoder-based models for anomaly detection. First, developing techniques to address the issue of over-confidence that AEs suffer from could be interesting. Also, it is reported that producing reliable uncertainties is still an ongoing research challenge [54], investigating production of reliable uncertainties would be interesting. Furthermore, in real-time anomaly detection scenarios, intrusion detection systems must operate on time-evolving data streams. Adapting the proposed approach under temporal settings would be interesting.

REFERENCES

- [1] M. Abdar et al., “A review of uncertainty quantification in deep learning: Techniques, applications and challenges,” *Inf. Fusion*, vol. 76, pp. 243–297, Dec. 2021.
- [2] K. Alsubhi, E. Al-Shaer, and R. Boutaba, “Alert prioritization in intrusion detection systems,” in *Proc. IEEE Netw. Oper. Manag. Symp. (NOMS)*, Apr. 2008, pp. 33–40.
- [3] M. E. Aminanto, T. Ban, R. Isawa, T. Takahashi, and D. Inoue, “Threat alert prioritization using isolation forest and stacked auto encoder with day-forward-chaining analysis,” *IEEE Access*, vol. 8, pp. 217977–217986, 2020.
- [4] F. Angiulli, F. Fassetti, and L. Ferragina, “Reconstruction error-based anomaly detection with few outlying examples,” 2023, *arXiv:2305.10464*.
- [5] P. Bergmann, M. Fauser, D. Sattlegger, and C. Steger, “A comprehensive real-world dataset for unsupervised anomaly detection,” in *Proc. IEEE/CVF Conf. Comput. Vision Pattern Recognit.*, Jun. 2019, pp. 9592–9600.
- [6] D. F. Brossart, V. C. Laird, and T. W. Armstrong, “Interpreting Kendall’s Tau and Tau-U for single-case experimental designs,” *Cogent Psychol.*, vol. 5, no. 1, Dec. 2018, Art. no. 1518687.
- [7] F. Cuppens, “Managing alerts in a multi-intrusion detection environment,” in *Proc. 17th Annu. Comput. Secur. Appl. Conf.*, 2001, pp. 22–31.
- [8] X. Fan, X. Zhang, and X. B. Yu, “Uncertainty quantification of a deep learning model for failure rate prediction of water distribution networks,” *Rel. Eng. Syst. Saf.*, vol. 236, Aug. 2023, Art. no. 109088.
- [9] D. Foata, “On the netto inversion number of a sequence,” *Proc. Amer. Math. Soc.*, vol. 19, no. 1, pp. 236–240, Feb. 1968.
- [10] G. M. Sullivan and R. Feinn, “Using effect size—Or why the P value is not enough,” *J. Graduate Med. Educ.*, vol. 4, no. 3, pp. 279–282, Sep. 2012.
- [11] Y. Gal and Z. Ghahramani, “Dropout as a Bayesian approximation: Representing model uncertainty in deep learning,” in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 1050–1059.
- [12] I. Golan and R. El-Yaniv, “Deep anomaly detection using geometric transformations,” in *Proc. Adv. Neural Inf. Process. Syst.*, 2018, pp. 1–12.
- [13] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proc. 34th Intl. Conf. Mach. Learn.*, Sydney, NSW, Australia, 2017, pp. 1321–1330.
- [14] J. A. Hoeting, D. Madigan, A. E. Raftery, and C. T. Volinsky, “Bayesian model averaging: A tutorial (with comments by M. Clyde, David draper and E. I. George, and a rejoinder by the authors),” *Stat. Sci.*, vol. 14, no. 4, pp. 382–417, Nov. 1999.
- [15] R. T. Ionescu, F. S. Khan, M.-I. Georgescu, and L. Shao, “Object-centric auto-encoders and dummy anomalies for abnormal event detection in video,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7842–7851.
- [16] Alex Kendall and Yarin Gal, “What uncertainties do we need in Bayesian deep learning for computer vision,” in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 1–11.
- [17] A. H. Khan, M. Hussain, and M. K. Malik, “ECG images dataset of cardiac and COVID-19 patients,” *Data Brief*, vol. 34, Feb. 2021, Art. no. 106762.
- [18] D. P. Kingma and M. Welling, “Auto-encoding variational Bayes,” 2013, *arXiv:1312.6114*.
- [19] R. Koenker and K. F. Hallock, “Quantile regression,” *J. Econ. Perspect.*, vol. 15, no. 4, pp. 143–156, 2001.

- [20] A. Krizhevsky and G. Hinton, "Learning multiple layers of features from tiny images," Univ. Toronto, Toronto, ON, Canada, Tech. Rep. TR-2009, 2009.
- [21] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 1–12.
- [22] Y. LeCun, C. Cortes, and C. Burges, "MNIST handwritten digit database," AT&T Labs, Florham Park, NJ, USA, Tech. Rep., 2010.
- [23] T. Li, Z. Wang, S. Liu, and W.-Y. Lin, "Deep unsupervised anomaly detection," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 3635–3644.
- [24] B. Lindemann, B. Maschler, N. Sahlab, and M. Weyrich, "A survey on anomaly detection for technical systems using LSTM networks," *Comput. Ind.*, vol. 131, Oct. 2021, Art. no. 103498.
- [25] Y. Liu, X. Shu, Y. Sun, J. Jang, and P. Mittal, "RAPID: Real-time alert investigation with context-aware prioritization for efficient threat discovery," in *Proc. 38th Annu. Comput. Secur. Appl. Conf.*, Dec. 2022, pp. 827–840.
- [26] J. F. Masakuna, K. N. D'Jeff, A. Soltani, M. Frappier, P.-M. Tardif, and F. Kabanza, "Streamlined and resource-efficient estimation of epistemic uncertainty in deep ensemble classification decision via regression," *IEEE Trans. Emerg. Topics in Comput. Intell.*, vol. 9, no. 4, p. 12, Aug. 2025.
- [27] J. F. Masakuna, S. W. Utete, and S. Kroon, "A coordinated search strategy for solitary robots," in *Proc. 3rd IEEE Int. Conf. Robotic Comput. (IRC)*, Feb. 2019, pp. 433–434.
- [28] J. F. Masakuna, S. W. Utete, and S. Kroon, "Performance-agnostic fusion of probabilistic classifier outputs," in *Proc. IEEE 23rd Int. Conf. Inf. Fusion (FUSION)*, Jul. 2020, pp. 1–8.
- [29] J. F. Masakuna and P. K. Kafunda, "Do prior information on performance of individual classifiers for fusion of probabilistic classifier outputs matter," *J. Classification*, vol. 40, no. 3, pp. 468–487, 2023.
- [30] Y. Mirsky, T. Doitshman, Y. Elovici, and A. Shabtai, "Kitsune: An ensemble of autoencoders for online network intrusion detection," in *Proc. Netw. Distrib. Syst. Secur. Symp.*, 2018, pp. 1–15.
- [31] S. M. Mohammad and L. Surya, "Security automation in information technology," *Int. J. Creative Res. Thoughts (IJCRT)*, vol. 6, pp. 901–905, Jun. 2018.
- [32] E. C. P. Neto, S. Dadkhah, R. Ferreira, A. Zohourian, R. Lu, and A. A. Ghorbani, "CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment," *Sensors*, vol. 23, no. 13, p. 5941, Jun. 2023.
- [33] J. Nixon, M. W. Dusenberry, L. Zhang, G. Jerfel, and D. Tran, "Measuring calibration in deep learning," in *Proc. CVPR Workshops*, vol. 2, 2019, pp. 38–41.
- [34] Y. Ovadia et al., "Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 1–12.
- [35] P. S. Palar, K. Zakaria, L. R. Zuhail, K. Shimoyama, and R. P. Liem, "Gaussian processes and support vector regression for uncertainty quantification in aerodynamics," in *Proc. AIAA Scitech Forum*, Jan. 2021, p. 181.
- [36] G. Pang, C. Shen, L. Cao, and A. Van Den Hengel, "Deep learning for anomaly detection: A review," *ACM Comput. Surv.*, vol. 54, no. 2, pp. 1–38, 2021.
- [37] A. Paszke et al., "Pytorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 1–12.
- [38] J. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," *Adv. Large Margin Classifiers*, vol. 10, no. 3, pp. 61–74, 1999.
- [39] A. A. Pol, V. Berger, C. Germain, G. Cerminara, and M. Pierini, "Anomaly detection with conditional variational autoencoders," in *Proc. 18th IEEE Int. Conf. Mach. Learn. Appl. (ICMLA)*, Dec. 2019, pp. 1651–1657.
- [40] C. Rasmussen, "The infinite Gaussian mixture model," in *Proc. Adv. Neural Inf. Process. Syst.*, 1999, pp. 1–7.
- [41] S. Sakazi, Y. Elovici, and A. Shabtai, "Prioritizing antivirus alerts on internal enterprise machines," in *Proc. Int. Conf. Detection Intrusions Malware, Vulnerability Assessment*. Cham, Switzerland: Springer, 2022, pp. 75–95.
- [42] H. Sarvari, C. Domeniconi, B. Prencak, and G. Stilo, "Unsupervised boosting-based autoencoder ensembles for outlier detection," in *Proc. 25th Pacific-Asia Conf. Adv. Knowl. Discovery Data Mining*. Cham, Switzerland: Springer, 2021, pp. 91–103.
- [43] T. Schlegl, P. Seeböck, S. M. Waldstein, G. Langs, and U. Schmidt-Erfurth, "F-AnoGAN: Fast unsupervised anomaly detection with generative adversarial networks," *Med. Image Anal.*, vol. 54, pp. 30–44, May 2019.
- [44] G. Shafer and V. Vovk, "A tutorial on conformal prediction," *J. Mach. Learn. Res.*, vol. 9, no. 3, pp. 371–421, 2008.
- [45] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward generating a new intrusion detection dataset and intrusion traffic characterization," in *Proc. 4th Int. Conf. Inf. Syst. Secur. Privacy*, 2018, pp. 108–116.
- [46] I. Sutskever, O. Vinyals, and Q. V. Le, "Sequence to sequence learning with neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2014, pp. 1–9.
- [47] M. Tavallaei, E. Bagheri, W. Lu, and A. A. Ghorbani, "A detailed analysis of the KDD CUP 99 data set," in *Proc. IEEE Symp. Comput. Intell. Secur. Defense Appl.*, Jul. 2009, pp. 1–6.
- [48] H. Torabi, S. L. Mirtaheri, and S. Greco, "Practical autoencoder based anomaly detection by using vector reconstruction error," *Cybersecurity*, vol. 6, no. 1, p. 1, 2023.
- [49] R. T. Ionescu, S. Smeureanu, B. Alexe, and M. Popescu, "Unmasking the abnormal events in video," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2895–2903.
- [50] R. Vallat, "Pingouin: Statistics in Python," *J. Open Source Softw.*, vol. 3, no. 31, p. 1026, Nov. 2018.
- [51] P. Virtanen et al., "SciPy 1.0: Fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, no. 3, pp. 261–272, 2020.
- [52] S. Warghade, S. Desai, and V. Patil, "Credit card fraud detection from imbalanced dataset using machine learning algorithm," *Int. J. Comput. Trends Technol.*, vol. 68, no. 3, pp. 22–28, Mar. 2020.
- [53] A. Yazdinejad, M. Kazemi, R. M. Parizi, A. Dehghantanha, and H. Karimipour, "An ensemble deep learning model for cyber threat hunting in industrial Internet of Things," *Digit. Commun. Netw.*, vol. 9, no. 1, pp. 101–110, Feb. 2023.
- [54] B. X. Yong and A. Brintrup, "Bayesian autoencoders with uncertainty quantification: Towards trustworthy anomaly detection," *Expert Syst. Appl.*, vol. 209, Dec. 2022, Art. no. 118196.
- [55] Z. Zeng, Z. Yang, D. Huang, and C.-J. Chung, "LICALITY—Likelihood and criticality: Vulnerability risk prioritization through logical reasoning and deep learning," *IEEE Trans. Netw. Service Manage.*, vol. 19, no. 2, pp. 1746–1760, Jun. 2022.
- [56] J. Zhai, S. Zhang, J. Chen, and Q. He, "Autoencoder and its various variants," in *Proc. IEEE Int. Conf. Syst., Man, Cybern. (SMC)*, Oct. 2018, pp. 415–419.
- [57] C. Zhong, J. Yen, P. Liu, and R. F. Erbacher, "Learning from experts' experience: Toward automated cyber security data triage," *IEEE Syst. J.*, vol. 13, no. 1, pp. 603–614, Mar. 2019.
- [58] B. Zong et al., "Deep autoencoding Gaussian mixture model for unsupervised anomaly detection," in *Proc. Int. Conf. Learn. Represent.*, 2018, pp. 1–19.
- [59] Y. Zou, J. Jeong, L. Pemula, D. Zhang, and O. Dabeer, "Spot-the-difference self-supervised pre-training for anomaly detection and segmentation," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 392–408.



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